CONTACT

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Loc: Calgary, Canada

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EDUCATION

M.Eng. Electrical & Computer
University of Calgary | 2024
GPA: 3.64 | Minor: Energy

B.Tech. Computer Science
REVA University | 2022
GPA: 8.7/10

SKILLS

Cloud: AWS, GCP, Firebase, Docker, K8s

Lang: Python, C++, Java, Dart, JS, Solidity

Web: Flutter, React, Angular, REST

Data: SQL, NoSQL, Big Data Pipelines

IoT: SCADA, PLCs, Modbus, Arduino

Quant: HFT, MT5, PineScript, Backtesting

AI: DeepSeek, LLMs, Prompt Eng.

Tools: Git, JIRA, Linux, Zena

CERTIFICATIONS

Blockchain Specialization (2020)

AWS Fundamentals (2020)

Data Science (Sybgen, 2020)

Google Cloud 30 Days (2020)

RAHUL JHA

Software Engineer | Industrial IoT & Quantitative Finance

PROFESSIONAL SUMMARY

Results-driven Software Engineer with dual expertise in Industrial Automation and Quantitative Finance. Proven track record of architecting full-stack IIoT ecosystems, optimizing high-frequency trading algorithms, and deploying scalable cloud architectures. Expert in bridging the gap between hardware sensors and advanced AI analytics to drive operational efficiency and revenue growth.

EXPERIENCE

Project Engineer

Litus | Calgary | May 2025 - Present

- Architected and deployed a full-stack Industrial IoT (IIoT) ecosystem, designing the company's first digital control system using PLCs and SCADA/HMI interfaces for remote automation.
- Engineered a high-throughput data pipeline orchestrating the synchronization of 500,000+ monthly data points from edge devices to cloud servers via Modbus TCP.
- Analyzed process data to identify bottlenecks, achieving a 30% improvement in lithium recovery efficiency by systematically optimizing pH, flow rates, and residence times.
- Spearheading the integration of AI-driven process control models and commercial automation systems for Nanomaterial Production, bridging the gap between pilot and mass manufacturing.
- Developed a proprietary software suite including custom plotting tools, internal ERPs, and management systems, reducing manual overhead for the R&D; team.
- Automated data logging infrastructure generating over 1,440 data points per instrument per day, eliminating manual entry and ensuring data integrity.
- Configured specialized test environments for advanced analytical equipment (Nanonord NMR), ensuring seamless API integration with central databases.

Lead Software Developer

TR PRO | Calgary | Feb 2025 - Present

- Designed and backtested a diverse portfolio of algorithmic trading strategies (HFT, Scalping, 'Down Trap Node'), optimizing execution logic for sub-millisecond latency.
- Pioneered advanced Prompt Engineering techniques to fine-tune DeepSeek LLMs, achieving >99% accuracy in parsing unstructured financial sentiment and signal data.
- Developed custom Expert Advisors (EAs) and automated trading bots within MetaTrader 5 (MT5), incorporating risk management layers to preserve capital during volatility.
- Architected the 'Overlord System,' a quantitative analysis engine validating trading signals in real-time via multi-timeframe (1M to 4H) market regime analysis.

- Led the migration to a modern Flutter/Flask architecture, engineering a multi-process Python backend managing multiple MT5 instances, increasing throughput by 500%.
- Engineered a dynamic rules engine for 8 distinct webhook strategies, mapping indicator strengths to optimal market regimes (Trending, Consolidation).
- Spearheaded creation of a user-friendly Flutter GUI and secure Firestore ecosystem, increasing engagement by 45% and reducing support tickets by 35%.

Software Engineer

FIS | Bengaluru | Feb 2022 - Sept 2022

- Maintained 99.5% accuracy in day-end processes for UAT and production, ensuring critical financial data integrity.
- Reduced system downtime by 20% through proactive troubleshooting and automated health-check scripts.
- Accelerated issue resolution times by 15% through optimization of incident response protocols.
- Achieved 100% SLA compliance for high-profile clients including RBC, BMO, PayPal, and American Express.
- Enhanced system capabilities by configuring new services in ZENA and establishing Big Data connections.

Application Developer

Codertech | Bengaluru | Aug 2020 - Sept 2022

- Engineered and launched cross-platform mobile apps (Android/iOS/Web) that scaled to support 60+ students in live classes, increasing user engagement and retention by 20%.
- Integrated in-app wallet and Razorpay payment gateway, boosting transaction success rates by 25%.
- Designed and implemented referral and commission systems, driving a 30% increase in user acquisition through network marketing.
- Optimized data flow with REST API communication between frontend and backend, delivering real-time quiz/exam functionality for groups.
- Elevated data security and reliability through Firebase and Cloud Firestore integration, safeguarding sensitive user and payment information.

Android Developer Intern

REVA NEST | Bengaluru | Feb 2020 - May 2020

- Engineered and launched cross-platform mobile apps (Android/iOS/Web) that scaled to support 60+ students in live classes, increasing user engagement and retention by 20%.
- Integrated in-app wallet and Razorpay payment gateway, boosting transaction success rates by 25%.
- Designed and implemented referral and commission systems, driving a 30% increase in user acquisition through network marketing.
- Optimized data flow with REST API communication between frontend and backend, delivering real-time quiz/exam functionality for groups.
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